3) HARD-CODED (OR GATE)

```
import numpy as np
x1 = np.array([0,0,1,1])
x2 = np.array([0,1,0,1])
t = np.array([0,1,1,1])
m = len(t)
max_iter = 4000
Learn_rate = 0.1
W1 = 0.1
W2 = 0.1
b = 0.0
def sigmoid (z):
    return 1/(1+np.exp(-z))
def sigmoid_derivative(s):
    return s*(1-s)
for _ in range(max_iter):
    dw1 = 0
    dw2 = 0
    db = 0
    for i in range(m):
        z = x1[i]*W1 + x2[i]*W2 + b
        y = sigmoid(z)
        error = t[i]-y
        dW1 += error * sigmoid_derivative(y) * x1[i]
        dW2 += error * sigmoid_derivative(y) * x2[i]
        db += error * sigmoid_derivative(y)
    W1 += Learn_rate *(dW1/m)
    W2 += Learn_rate *(dW2/m)
    b += Learn_rate *(db/m)
print("W1:", W1, "W2:", W2, "b:", b)
final_z = x1*W1 + x2*W2 + b
prediction = sigmoid(final_z)
print("prediction:", prediction.round())
```

OUTPUT

```
W1: 3.3199009616766952
W2: 3.3199009616766952
b: -1.3513760666009
prediction: [0. 1. 1. 1.]
```